Not the Metric You Report: Leakage-Audited Landmine Surface Detection, and Why mAP_{50} Cannot Choose an Operating Point

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Abstract

Detection benchmarks in aerial and remote-sensing settings are usually ranked by mean average precision (mAP). We show that on a small, single-class landmine surface-indicator dataset this ranking is uninformative for deployment: a 3.0 M-parameter detector and an 11.1 M-parameter detector differ by less than the seed-to-seed spread in test mAP_{50} (0.979 vs. 0.977 ± 0.003), yet their recall-first operating points differ by a factor of 4.3 in false positives per image (6.50 vs. 1.51). We argue that the meaningful object of study is the operating point, not the summary curve, and we build the evaluation protocol around that claim.

The study rests on two methodological contributions that require no new architecture. First, we audit a widely reused landmine image corpus and find that its published split leaks: 76 Roboflow base-name groups span splits and 63.8% of evaluation images have a near-duplicate in training, consistent with augmentation applied before splitting. We rebuild the split group-wise, apply a two-method leak filter (perceptual hash plus pixel MSE) with manual review of every candidate, and remove 30 evaluation images. The frozen manifest (3,907 / 181 / 189 images; digest `748322d6...`) contains no exact duplicate, no base-key collision, and no confirmed near-duplicate leak.

Second, we measure what the leak was worth. A detector trained on the leaky split scores 0.994 mAP_{50} on the full clean test set against 0.974 for a clean-trained model, but on the 49 contamination-free test images the two are tied (0.995 vs. 0.995). The apparent inflation is therefore contamination, not capability—and it is small, a negative result we report rather than bury.

We then evaluate a MobileNetV3-Large second stage over detector crops on three detector seeds. The two-stage arm improves precision over a recall-matched single-stage baseline at two of three seeds ($\Delta P = +0.128$ and $+0.144$, bootstrap intervals excluding zero) but fails at the third ($\Delta P = -0.037$, interval $[-0.077, +0.013]$). Under a max-F_2 rule a val-tuned single-stage threshold beats the two-stage pipeline outright ($P = 0.976$ vs. 0.945). We report the effect as detector-dependent rather than architectural, and we identify the mechanism: the two-stage gain comes from rescuing seeds whose val-selected threshold transferred poorly, not from better localisation. Grad-CAM confirms that the detector's evidence lies on the objects (median $58\times$ energy enrichment over chance) and that the second stage adds almost no spatial evidence of its own ($1.82\times$ for the image-level baseline, $2.13\times$ for crops)—a caution against reading a classifier's accuracy as reasoning. All experiments were run on one 8 GB laptop GPU within a $\sim$30 h budget; we state explicitly what that budget forbade.

1 Introduction

A landmine detector that is evaluated with mean average precision is being asked the wrong question. Mean average precision integrates performance over every confidence threshold and every recall level, then reduces the result to one number. A demining team does not operate a curve; it operates one threshold, on one day, with one tolerance for false alarms. Two systems can share a curve and behave completely differently at the threshold that will actually be used. This paper is an extended argument, with measurements, that the operating point—and not the summary curve—is the object that should be reported, and that the usually unglamorous work of auditing the split is what makes any such comparison believable.

We study single-class detection of *surface* landmine indicators from downward-looking aerial imagery. The scope limitation is physical and we state it first: a visible-light or colour camera cannot see a buried object. What it can see is a mine lying on the surface, a partially exposed casing, or a surface marker. Claims in this paper are therefore claims about visible surface objects, not about buried ordnance, and nothing here is a field-validated clearance system. The task is nonetheless a useful stress test for detection methodology, because the class is rare, the imagery is heterogeneous, and the published data reuse is heavy.

What we found. Three results organise the paper.

1. **The split leaks, and the leak is measurable.** The combined landmine corpus contains 76 filename groups (Roboflow base names) that span train and evaluation splits, and 63.8% of validation and test images have a perceptual-hash neighbour in training. This is the signature of augmentation applied before splitting: the same scene, re-exported, lands on both sides of the boundary. After rebuilding the split group-wise and filtering twice, 30 evaluation images are removed as true leaks. A detector trained on the leaky split scores 0.994 mAP_{50} on the full clean test set, against 0.974 for a clean-trained detector. On the 49 test images that are contamination-free for both, the two are tied at 0.995 and 0.995. The leak inflated the number, but by less than the folklore predicts—a negative result, and one that would have been invisible without the contamination-free subset.

2. **mAP_{50} saturates before the deployment question is asked.** A YOLOv8s detector over three seeds reaches 0.977 ± 0.003 test mAP_{50}; a YOLOv8n detector with 3.7× fewer parameters reaches 0.979. Ranked by mAP, the smaller model is as good or better, at 8.2 GFLOPs instead of 28.6. Ranked by the recall-first operating point chosen on validation, the smaller model produces 6.50 false positives per image against 1.51—a 4.3× difference that no mAP column reveals. The ranking reversal is our headline finding, and it is a statement about evaluation, not about architectures.

3. **The two-stage gain is detector-dependent, and we can say why.** A MobileNetV3-Large classifier over detector crops improves precision over a recall-matched single-stage baseline at two of three detector seeds ($\Delta P = +0.128$ and $+0.144$, paired bootstrap intervals excluding zero). At the third seed it is *worse* ($\Delta P = -0.037$, interval $[-0.077, +0.013]$). Under a max-F_2 rule, a single confidence threshold tuned on validation beats the two-stage pipeline outright ($P = 0.976$ vs. 0.945). The mechanism is not that the classifier localises better. It is that the classifier rescues seeds whose validation-selected threshold transferred poorly to test. Where the baseline threshold transfers well, the second stage has nothing to add. Reporting a three-seed mean of 0.879 ± 0.018 would have hidden this entirely; we report the per-seed structure instead.

Why this is worth publishing. None of the three findings required a new architecture, a large GPU, or a new dataset. They required (i) reconstructing the split from scratch, (ii) refusing to average over seeds when the seeds disagree structurally, and (iii) measuring at the operating point rather than at the summary. The total compute was one 8 GB laptop GPU and roughly 30 hours. We think this is a realistic and under-reported budget for applied detection work, and we document what it forbade (Section 9) rather than implying broader coverage.

Contributions.

- A reproducible leakage audit of a publicly derived landmine corpus: an operational leak definition, a two-method filter with a manually validated MSE cut, a frozen manifest, and an accounting of all 30 removals (Section 4).
- A direct leaky-versus-clean comparison on a contamination-free subset, including the unfavourable result that the leak's effect on mAP_{50} is small (Section 4.4).
- An operating-point evaluation protocol—thresholds chosen on validation only, applied once to test, with paired image bootstrap and source-cluster intervals—and the demonstration that it reverses an mAP-based ranking (Section 7).
- A per-seed analysis of a two-stage detector-plus-classifier pipeline that localises a failure rather than averaging it away, together with a max-F_2 analysis showing when the second stage is not worth its cost (Section 7.4).
- A Grad-CAM audit used as a sanity check on evidence, with an explicit separation between attribution and correctness, showing that the image-level classifier's accuracy is not accompanied by object-level evidence (Section 7.6).

Section 2 positions the work; Section 3 describes the corpus and its provenance; Section 4 the audit; Sections 5–6 the models and protocol; Section 7 the results; Sections 8–10 discussion, limitations and ethics. Code, weights, tables and the frozen split manifest accompany the paper (Section 11).

2 Related Work

Landmine and UXO detection. The physical sensing literature is large and predates deep learning by decades: metal detectors, ground-penetrating radar, infrared and multispectral sensing, and combinations thereof (Siegel, 2002; Kasban et al., 2010). What matters for this paper is a distinction that the computer-vision literature often blurs. Subsurface sensing asks whether a buried object is present; visible-band aerial imaging asks whether a surface object or marker is visible. The two tasks have different base rates, different failure modes, and different false-alarm costs. We therefore restrict every claim to the visible-surface task and treat buried-object performance as out of scope rather than as a gap to be estimated. The humanitarian scale of the problem is documented annually by the Landmine Monitor (International Campaign to Ban Landmines – Cluster Munition Coalition, 2023); we mention it to motivate careful evaluation, not to claim operational utility.

Detectors. The YOLO family (Redmon and Farhadi, 2018; Jocher et al., 2023) remains the default for small-object aerial detection because of its latency at moderate resolution. Alternatives include anchor-free and transformer-based detectors (Tian et al., 2019; Carion et al., 2020; Zhao et al., 2024) and two-stage region proposals (Ren et al., 2015), with focal loss (Lin et al., 2017) addressing

foreground imbalance. Our contribution is not a detector. We deliberately use the most standard available baseline, because the point is that evaluation choices dominate architecture choices at this scale. The lightweight classifier in our second stage is MobileNetV3 (Howard et al., 2019), chosen for cost rather than novelty.

Leakage and reproducibility. That machine-learning results can be inflated by information leaking across train and evaluation splits is now well documented, both as a general methodological failure (Kapoor and Narayanan, 2023; Kapoor et al., 2024; Roberts et al., 2021) and in specific benchmarks (Recht et al., 2019; Barz and Denzler, 2020). The canonical defence is group-aware splitting, so that no group—scene, patient, site, or source document—spans the boundary (Kohavi, 1995). In image corpora, groups must be recovered from pixels when filenames are unreliable: perceptual hashing (Zauner, 2010) proposes near-duplicate candidates, and a pixel-space distance such as MSE or SSIM (Wang et al., 2004) separates true re-exports from terrain-texture coincidences. Our audit follows exactly this two-stage pattern, and we report the threshold at which the two methods disagree. Publicly shared detection datasets frequently originate as re-exports of the same source collection, which is why base-key collisions matter independently of pixels.

Evaluation beyond mAP. The mismatch between average precision and deployed behaviour is an old observation in detection. Hoiem et al. (2012) argued for error diagnosis over single scores; Dollár et al. (2012) popularised log-average miss rate at a fixed false-positive-per-image budget precisely because average precision hides the low-false-alarm regime. Our protocol adopts false-positives-per-image at a recall-first operating point as the primary quantity, and reports mAP only as a secondary, comparable-to-the-literature number. We also follow the recommendation to report intervals rather than point estimates: paired image bootstrap for comparisons on the same images (Efron and Tibshirani, 1993), and source-cluster resampling where the effective sample size is the number of acquisition sources, not the number of images. Agreement statistics use Cohen's κ (Cohen, 1960) for image-level decisions. Dataset and model documentation follow the datasheet and model-card conventions (Gebru et al., 2021; Mitchell et al., 2019).

Attribution methods. Grad-CAM (Selvaraju et al., 2017) and its variants (Chattopadhyay et al., 2018) are widely used to argue that a model "looks at the right thing". The evidence that saliency maps can be insensitive to model parameters (Adebayo et al., 2018), and that deletion-style tests are the appropriate falsification (Samek et al., 2017), means attribution should be treated as a diagnostic and never as proof of correctness. The Clever Hans literature (Lapuschkin et al., 2019) is the cautionary case: a classifier can be accurate for the wrong reason. We use Grad-CAM in that falsification role only, and we report where it fails.

3 Data

3.1 Corpus and task

The corpus pools five public Roboflow Universe landmine-detection exports, identified here by source id: `ds0`, `ds1`, `May_lwir`, `Jan_lwir`, and a small derived set folded into `ds0`. Each image carries one class, `landmine`, and axis-aligned boxes. There is no mine-type label; the task is presence and localisation of a visible mine-like object. Table 1 gives the frozen split.

The class is heavily imbalanced in the detection sense and, more importantly, heterogeneous in acquisition. Table 2 shows the source/split shift that dominates the results later. `Jan_lwir` images

Table 1: Frozen dataset `landmine_combined_v2`. "neg" counts images with an empty label file (deliberate negatives), not missing labels; there are zero missing labels. Manifest digest `748322d6...c80de`.

split	images	instances	neg	sources
train	3,907	11,550	20	Jan 100, May 776, ds0 2,589, ds1 442
valid	181	419	7	Jan 42, ds0 138, ds1 1
test	189	253	5	Jan 10, ds0 151, ds1 28

Table 2: Source and split composition. The test set is dominated by `ds0` images (151 of 189) but by `Jan` boxes (77 of 253).

split/source		images	boxes	boxes/img	empty rate
test	Jan_lwir	10	77	7.70	0.00%
test	ds0	151	148	0.98	1.99%
test	ds1	28	28	1.00	7.14%
train	Jan_lwir	100	740	7.40	0.00%
train	May_lwir	776	7,706	9.93	0.00%
train	ds0	2,589	2,643	1.02	0.77%
train	ds1	442	461	1.04	0.00%
valid	Jan_lwir	42	287	6.83	0.00%
valid	ds0	138	131	0.95	5.07%
valid	ds1	1	1	1.00	0.00%

average 7.7 boxes per test image; `ds0` and `ds1` average about one. A model that learns "many objects" from `Jan` and applies it to `ds0` will generate false positives by construction, and vice versa. This is not a nuisance to be averaged away: it is the principal source of variance in the test set, and we return to it in Section 7.2.

3.2 Two content defects, disclosed

The `lwir` files are not thermal. Sources named `May_lwir` and `Jan_lwir` suggest long-wave infrared imagery. They are not. Mean HSV saturation on a 24-image sample is 0.38, against 0.25 for the `ds0` RGB reference, and the images carry visible marker pins. They are colour RGB photographs. We keep them, because removing them would drop the only multi-object source and make the task artificially easy, and we disclose the consequence: a model could exploit marker pins as a shortcut. A twelve-crop zoom check around test mines found no consistent marker, paint, or flag cue, but we cannot rule out a subtler one, and we do not make thermal claims anywhere in this paper.

Two test labels are objectively wrong. Manual review identified two test boxes that are not on mines: `ds0_0801` annotates a tree trunk and `ds0_0883` annotates a fence. Both were confirmed independently by a Grad-CAM energy of exactly zero (Section 7.6). We remove them from ground truth and report results both with and without the correction (Table 3). No training labels were found to be wrong, so no retraining was performed.

3.3 Provenance and redistribution

The source archives were obtained as public exports, but their licence terms were not recorded at download time and the original project URLs are no longer available to this project. We therefore do

Table 3: Effect of removing two objectively wrong test boxes (no retraining). Intervals in the source artifact are unchanged; the shift is a ground-truth correction, not a model change.

arm	metric	before	after
C (two-stage)	box recall	0.9644	0.9721
C (two-stage)	image recall	0.9891	1.0000
C (two-stage)	FP/image	0.153	0.153
A (YOLO deploy)	box recall	0.9763	0.9841
B (YOLO matched)	box recall	0.9723	0.9801

not redistribute any image, label file, or crop. What is released is the split manifest with a SHA-256 digest, the removed-leak list with reasons, the audit CSVs and montages, the trained weights, and the code. Anyone holding the original archives can reconstruct the exact splits byte-for-byte from the manifest. Full detail is in `DATA_PROVENANCE.md`; the weights carry the upstream Ultralytics AGPL-3.0 terms (Section 11). We regard the missing licence record as a defect of this study and state it rather than inferring a permissive licence.

4 Leakage Audit

4.1 The original split fails

The first audit pass over the pooled corpus returned a clear failure. Of the evaluation images, 63.8% had a training image within perceptual-hash Hamming distance 2, and 76 Roboflow base names appeared in more than one split (train∩valid 37, train∩test 31, valid∩test 10). A montage of the base-name collisions shows the same physical scene, the same mine placement, and different export bytes: these are re-exports, not distinct photographs. The mechanism is augmentation before splitting, present in the upstream generation scripts. The test split also contained zero negative images, so false-alarm rate was undefined. We therefore discarded the split entirely rather than attempting to patch it.

4.2 Operational leak definition

A validation or test image x is a leak if any training image shares any of:

1. the exact MD5 digest;
2. the Roboflow base key (filename with the `*.rf.<md5>.jpg` export suffix stripped);
3. the same source id *and* a 64-bit perceptual hash (Zauner, 2010) within Hamming distance 2 under all eight dihedral transforms, *and* a 32×32 RGB mean-squared error below 900.

Condition 3 is a conjunction for a reason. Perceptual hash alone over-fires on repetitive terrain: grass and dirt produce Hamming collisions that are not the same scene. The MSE term separates the two populations, and manual review sets the cut. Every Hamming-≤ 2 candidate—146 pairs across two rounds—was rendered side by side and inspected. Confirmed same-scene pairs occurred at MSE ≤ 817; terrain-texture coincidences at MSE ≥ 958.6. The gap is narrow but unambiguous on montage, and the cut at 900 sits inside it.

Table 4: Leakage audit outcome for the frozen split. "Residual candidates" are Hamming-≤ 2 pairs rejected by the pixel-MSE test and manual review.

quantity	value
Images (train / valid / test)	3,907 / 181 / 189
Instances	11,550 / 419 / 253
Negatives	20 / 7 / 5
Exact cross-split duplicate groups	0
Base-key collisions (train-valid / train-test / valid-test)	0 / 0 / 0
Residual near-duplicate candidates	99 (all MSE $\geq$ 958.6)
Evaluation images removed as leaks	30 (19 valid, 11 test)
Confirmed true leak rate	$0/370 = 0.0\%$
Manifest SHA-256	`748322d6...c80de`

Table 5: Leaky-versus-clean comparison. The leaky model's operating confidence (0.065) was chosen on the old validation split; the clean model's (0.02) on the new one. P and FP/image are therefore not threshold-matched and mAP is the fair column.

model	evaluation set	mAP_{50}	$mAP_{50–95}$	P	FP/img
leaky v1	v1 test (own split)	0.9697	0.6436	0.834	n/a (0 neg)
leaky v1	v2 test, full (74.1% contaminated)	0.9940	—	0.857	0.222
clean v2 s0	v2 test, full	0.9741	0.6924	0.427	1.751
leaky v1	v2 test, contamination-free ($n = 49$)	0.9950	0.7250	0.863	0.204
clean v2 s0	v2 test, contamination-free ($n = 49$)	0.9947	0.7010	0.354	2.347

4.3 Rebuilding and freezing

Groups were formed from base key, capture session, and same-source hash edges, yielding 1,048 groups, then assigned whole to a split. Two filtering passes removed 30 evaluation images: 13 at MSE ≤ 100, then 17 more at MSE ≤ 900, including a cross-folder duplicate (`ds0_0043`↔`ds1_0043`). The split is 19 validation and 11 test images lighter as a result. The residual 99 of 370 evaluation images with a Hamming-≤ 2 training neighbour all have MSE ≥ 958.6 and were verified as texture false positives. True leak rate: 0/370.

4.4 What the leak was worth

An audit is only useful if it changes something. We therefore trained a YOLOv8s detector on the *old* split (the "leaky" model) and compared it with the clean-trained seed-0 detector on the clean test set. This is the natural before/after experiment, and it has a trap: 140 of 189 clean test images (74.1%) share a v2 group with an image the leaky model trained on. Scoring the leaky model on the full test set therefore measures memorisation.

The results are in Table 5. On its own split the leaky model reaches 0.970 mAP_{50}, but that split has zero negatives, so false-positive rate is undefined and the number cannot be compared with anything in this paper. On the full clean test set it reaches 0.994, comfortably above the clean model's 0.974. On the 49 contamination-free images the two are tied: 0.9950 against 0.9947 mAP_{50}, and 0.725 against 0.701 $mAP_{50–95}$. The 49-image subset is small and the comparison is therefore low-powered, but the direction is unambiguous: the apparent +0.020 advantage is contamination, and the honest estimate of the leak's effect on this corpus is close to zero.

This is a negative result and we treat it as one. The literature on leakage (Kapoor and Narayanan,

2023; Recht et al., 2019) prepares one to expect a large inflation. Here the clean and leaky models converge on the leak-free images, which suggests that for this task the leaked material mostly taught the model the same thing the clean data does. That is not a reason to skip the audit—the old split remains unusable, because its selection split was contaminated and its test split had no negatives—but it is a reason to report the size of the effect honestly rather than using the audit as a rhetorical amplifier. It also has a practical implication: an audit that only reports "we fixed leakage" without measuring the contamination-free delta leaves the reader unable to judge whether the fix mattered.

5 Method

5.1 Detector

We use Ultralytics YOLOv8 (Jocher et al., 2023) at two depths, `yolov8s` and `yolov8n`, both with a single-class head. Models are initialised from COCO-pretrained weights and fine-tuned for 75 epochs at 640×640, batch 8, with the Ultralytics default augmentation recipe and early-stopping patience 20. The `s` model is trained with seeds $0, 1, 2$; the `n` model with seed 0 only, as a single size point. All runs use the same pinned stack (Ultralytics 8.4.117, PyTorch 2.6.0+cu124) on one RTX 4070 Laptop GPU with 8 GB of memory.

Checkpoint selection is done on validation. For each seed we keep the checkpoint (last or best) that gives the better validation operating point, then choose the confidence threshold on validation and apply it unchanged to test. No test quantity influences any choice. This is the only point in the protocol where discipline costs something: for some seeds the validation-optimal threshold transfers badly to test, and we do not fix that after the fact (Section 7.4).

5.2 Operating points

For a set of detections, precision P, recall R and false positives per image FP/img are all functions of a confidence threshold. We define two operating points on validation and carry both to test:

- **Recall-first (OP_{97}).** The largest confidence whose validation box recall is at least 0.97. This is the operating point a demining workflow would plausibly choose: a missed mine is a safety event, a false alarm is an analyst cost. It is the primary point in all tables.
- **High-precision (OP_{95}).** The largest confidence with validation recall at least 0.95, used only for the image-level comparison.

mAP is computed in the usual way at a low confidence floor (0.001) and reported for comparability with the literature. It plays no role in any selection.

5.3 Two-stage pipeline

The second stage is a MobileNetV3-Large classifier (Howard et al., 2019) applied to detector crops. Each detection box is expanded by 20% on each side, cropped, and resized to 224×224; the classifier outputs a single sigmoid score. Training crops are 11,550 positives and 15,411 negatives, the latter comprising random background, 3,730 detector false positives mined at confidence 0.05 from an epoch-20 checkpoint, and 20 empty-frame samples. Positives and negatives are sampled 1:1 per epoch; the classifier is fine-tuned end-to-end for 30 epochs at learning rate 2×10^{-5} with per-sample class weights $[97.7, 0.50]$ to compensate the 3,887:20 image-level imbalance. The 1:1 ratio and the

unfrozen-late-layers recipe were chosen on validation, and a five-variant ablation is reported in Section 7.5.

We compare three arms, all evaluated on test exactly once with validation-frozen parameters:

- **A (deploy).** YOLO at its validation-chosen OP_{97} confidence.
- **B (matched recall).** YOLO at the confidence that matches arm A's *validation* recall. This isolates the classifier's contribution from the trivial effect of thresholding harder.
- **C (two-stage).** YOLO proposals at a fixed low confidence (0.05) are routed through the classifier; a proposal is kept if it passes either a *protection* rule (detector confidence $\geq P$) or the classifier threshold τ. P is swept over $\{0.4, 0.6, 0.8, \infty\}$ on validation; for each P we take the largest τ whose validation recall is within 0.5 points of stage-1 recall, then choose the pair with the lowest validation FP/image. All three seeds select $P = 0.4$, with $\tau = 0.27/0.23/0.22$ for seeds 0/1/2.

The protection rule matters. Without it the classifier can veto a high-confidence detection, and on this corpus that costs recall on the multi-object `Jan` source. With $P = 0.4$ the pipeline is allowed to suppress only low-confidence proposals, which is the regime where the classifier's crop evidence is useful.

5.4 Statistics

All test comparisons use the same 189 images, so we resample images, not detections. For a paired comparison between two arms we draw 5,000 bootstrap replicates of the image set (Efron and Tibshirani, 1993) and report the 95% percentile interval of the difference. For single-arm uncertainty we additionally report a source-cluster bootstrap over the three test sources, and a leave-one-source-out analysis; with three clusters these intervals are wide by construction, and we present them as such rather than as precise estimates. Image-level agreement uses Cohen's κ (Cohen, 1960) alongside balanced accuracy, because the test set has only 5 negative images and raw accuracy is dominated by the positive class.

5.5 Attribution audit

We compute Grad-CAM (Selvaraju et al., 2017) on the neck feature maps feeding the detection head, targeting the top pre-NMS class score, and on the last convolutional block of the classifier. Because the top pre-NMS anchor usually sits on the stride-8 head, the largest stride with a non-zero gradient is used; this deviates from the common stride-32 choice and is documented with the code. We report four quantities: the share of CAM energy inside the ground-truth box (and its enrichment over the box's area share), whether the CAM peak falls inside the box, the drop in detection score when the box is masked versus when a matched background region is masked, and a head-to-head comparison of detector and classifier saliency on the same crops. Every sentence about these numbers is phrased as evidence about *where the model looked*, never about whether the prediction is correct.

6 Experimental Protocol and Reproducibility

Frozen rules. Five rules governed every run, and we report violations as well as compliance. (i) Thresholds are chosen on validation only. (ii) Every evaluation that touches test is logged, numbered, and justified in advance. (iii) Every number in this paper is produced by a committed script reading a committed JSON or CSV. (iv) The original dataset and the pre-audit runs are never modified. (v) Test labels are corrected only when objectively wrong, with before/after numbers reported.

Test-touch accounting. The full log contains 18 numbered touches. Four are large: the seed-0 frozen-protocol detector evaluation, the seed-0 and seed-1/2 two-stage evaluations, and the corrected detector-aligned two-stage re-scoring. The remainder are the per-seed detector evaluations, the MNv3 baseline evaluations, and the Grad-CAM passes, each sanctioned by the protocol before execution. We also disclose a deviation: the upstream training script auto-evaluates test at confidence 0.25 at the end of training, and this behaviour could not be disabled by flag. It is recorded as D017. No threshold, checkpoint, or architecture choice references those numbers; they appear in no table here. The frozen-protocol evaluation is a separate, single pass.

Correction we had to make. An earlier version of the three-seed two-stage result reused the seed-0 detector for all three seeds, producing a tight mean of 0.879 ± 0.018. The arms A and B records were byte-identical across seeds, which is the tell. Re-running the pipeline with each seed's own detector gives 0.894/0.825/0.724 and a standard deviation of 0.086. The tight interval was an artifact of a shared component. We report the corrected numbers throughout and flag the earlier figure as withdrawn.

Compute. All training and evaluation ran on one NVIDIA RTX 4070 Laptop GPU with 8 GB of memory, under Windows, within an approved budget of roughly 30 GPU-hours. Detector training took 50–85 minutes per seed; the second-stage classifier 30 epochs per run; the ablation matrix five classifier retrains. Evaluation uses 640×640 input, NMS IoU 0.7, `max_det` 300, FP32, no test-time augmentation. Latency figures are PC GPU FP32 medians over 50 repetitions after 10 warm-up iterations, and are *not* edge-device numbers; no Jetson or TensorRT measurement was performed.

What the budget forbade. We did not train YOLOv8m or YOLOv8l on the clean split, did not evaluate a second detector family (RT-DETR or Faster R-CNN), did not run grouped cross-validation, did not re-mine hard negatives on data the detector had not seen, and did not retrain after the test-label correction. Each omission is a real limitation and is listed in Section 9. We prefer to name them than to let a single-seed or single-family result stand in for a claim it cannot support.

Artifacts. The repository contains the code, the six detector checkpoints (three seeds, ONNX included), the three stage-2 checkpoints, the frozen split manifest and its digest, all evaluation JSONs, the generated tables, and the figures. Regenerating the tables requires no GPU and no dataset: the committed JSONs are sufficient, and the table script fails loudly if a required field is missing rather than silently substituting a value.

7 Results

7.1 Size sweep: mAP saturates, the operating point does not

Table 6 reports both depths. The mAP columns are almost flat: 0.977 ± 0.003 for `yolov8s` over three seeds and 0.979 for `yolov8n` at one seed. On validation the gap is larger (0.954 vs. 0.932), which means we cannot claim the smaller model is better on this evidence: the difference is inside the seed spread of the larger model, and a single-seed estimate on 189 test images cannot resolve it. The defensible reading is that mAP_{50} cannot distinguish the two detectors.

The operating-point columns are not flat at all. At the recall-first point the smaller detector emits 6.50 false positives per image against 1.51 for the larger one—a $4.3\times$ difference, and notice the direction: the model with fewer parameters is *worse* at deployment while being equal or better on

Table 6: Size sweep on the frozen test split. `s` is mean ± std over seeds 0/1/2; `n` is seed 0 only and therefore carries no interval. "OP" is the validation-chosen recall-first operating point ($R \geq 0.97$ on validation), applied unchanged to test. Latency is PC GPU FP32, 640×640, median of 50 runs.

model	params	GFLOPs	MB	ms	val mAP_{50}	test mAP_{50}	R	P	FP/img
yolov8s	11.1 M	28.6	21.5	6.5	0.9544 ± 0.0027	0.9768 ± 0.0034	0.9763 ± 0.0040	0.5157 ± 0.2180	1.510 ± 1.009
yolov8n	3.0 M	8.2	6.0	6.9	0.9322	0.9787	0.9842	0.1686	6.497

Table 7: Per-seed operating points for `yolov8s`. The threshold chosen on validation varies by an order of magnitude across seeds (0.01 to 0.21) and so does the resulting precision. Test mAP barely moves.

seed	val conf	test mAP_{50}	box P	box R	FP/img	image R [95% CI]
0	0.020	0.9741	0.427	0.976	1.751	1.000 [0.980, 1.000]
1	0.010	0.9807	0.356	0.980	2.376	0.995 [0.970, 0.999]
2	0.210	0.9756	0.764	0.972	0.402	0.984 [0.953, 0.994]

mAP. Its precision at that point is 0.169 against 0.516. The mAP_{50} of 0.979 is earned by ranking, not by calibration, and at a fixed high-recall threshold the ranking advantage evaporates.

This is the paper's central empirical claim, and it is a claim about evaluation. A benchmark that reports only mAP_{50} would rank `yolov8n` first and recommend it for a 8.2 GFLOP budget; a deployment engineer would get a detector emitting four times as many alarms per frame. We note one more inconvenient detail: measured latency does not favour the smaller model (6.9 vs. 6.5 ms), because at this size the fixed per-image overhead dominates. The GFLOP and parameter savings are real, the wall-clock saving on this hardware is not. Reporting FLOPs as a proxy for latency is another summary statistic that fails to transfer.

Table 7 exposes the mechanism. The three seeds reach nearly identical test mAP (0.974–0.981) and nearly identical recall (0.972–0.980), but the validation procedure selects thresholds of 0.02, 0.01, and 0.21, producing precisions of 0.427, 0.356, and 0.764. Seed 2 is eight to ten times more conservative at the same mAP. Any claim about "the model's precision" that pools these seeds is a claim about the threshold-selection procedure, not about the detector. We therefore report per-seed values wherever precision or FP/image is the quantity of interest, and treat the three-seed mean as a summary of protocol variance rather than of model quality.

7.2 Where the variance lives: source structure

The source-cluster bootstrap makes the same point from a different direction. For seed 0, resampling the three test sources gives $P \in [0.195, 0.874]$, $R \in [0.948, 1.000]$, and FP/img $\in [0.139, 30.1]$. With three clusters these intervals are honest but almost uninformative; the leave-one-source-out analysis is more legible. Removing `Jan_lwir`—the ten images that generate most of the false alarms—raises seed-0 precision from 0.427 to 0.853 and drops FP/image from 1.75 to 0.17. Removing `ds0` does the opposite: precision falls to 0.246 and FP/image rises to 8.16, because the remaining `Jan` false positives are divided by far fewer true positives.

Table 8 gives the direct decomposition. Ten `Jan` test images account for 301 of the 331 false positives emitted by arm A—91% of all false alarms come from 5% of the images. The two-stage arm reduces that to 21, and reduces total FP/image from 1.75 to 0.15, but it also costs recall on that source ($0.948 \rightarrow 0.909$). The honest summary is that the aggregate metric is dominated by a source that is over-represented in boxes and under-represented in images, and that any intervention which suppresses `Jan` false alarms will look like a large aggregate gain while being a modest, source-specific

Table 8: Per-source test results for seeds 0, arm A (YOLO at deploy threshold) and arm C (two-stage). All 331 arm-A false positives come from three sources; 301 of them from ten `Jan` images.

arm	source	images	TP	FP	P	R	FP/img
A	Jan_lwir	10	73	301	0.195	0.948	30.100
A	ds0	151	146	21	0.874	0.986	0.139
A	ds1	28	28	9	0.757	1.000	0.321
A	all	189	247	331	0.427	0.976	1.751
C	Jan_lwir	10	70	21	0.769	0.909	2.100
C	ds0	151	146	4	0.973	0.986	0.026
C	ds1	28	28	4	0.875	1.000	0.143
C	all	189	244	29	0.894	0.964	0.153

Table 9: Image-level decisions on test. MNv3 A is seed 0; A+B is mean ± std over three seeds; YOLO and two-stage rows are mean ± std over three detector seeds. Balanced accuracy and κ are the meaningful columns given only five negative images.

system	accuracy	balanced acc	precision	recall	κ
MNv3 A (thr 0.5)	0.974	0.500	0.974	1.000	0.000
MNv3 A (recall-first)	0.968	0.497	0.973	0.995	−0.009
MNv3 A+B (thr 0.5)	0.974 ± 0.000	0.500 ± 0.000	0.974 ± 0.000	1.000 ± 0.000	0.000 ± 0.000
MNv3 A+B (recall-first)	0.968 ± 0.005	0.562 ± 0.056	0.977 ± 0.003	0.991 ± 0.006	0.156 ± 0.148
YOLO deploy	0.984 ± 0.009	0.830 ± 0.115	0.991 ± 0.006	0.993 ± 0.008	0.677 ± 0.199
YOLO OP_{95}	0.986 ± 0.003	0.993 ± 0.002	1.000 ± 0.000	0.986 ± 0.003	0.784 ± 0.038
Two-stage (arm C)	0.989 ± 0.005	0.962 ± 0.059	0.998 ± 0.003	0.991 ± 0.003	0.818 ± 0.094

change.

We flag a confound: `Jan` and `May` are the sources whose filenames claim thermal imagery but whose pixels are colour photographs with marker pins (Section 3). If a marker cue is present, `Jan` false alarms may be marker detections rather than mine detections, and the second stage may be suppressing markers rather than terrain. Our twelve-crop check did not find a consistent marker near test mines, but the confound cannot be excluded from this data, and it is one reason we do not claim the two-stage gain is about mines.

7.3 Image level

Table 9 asks a coarser question: does the image contain a mine at all? The frozen-feature MobileNetV3 baseline is degenerate on this task. Its stage-A variant reaches 0.974 accuracy but balanced accuracy 0.500 and $\kappa = 0.000$: it calls every image positive, which is correct for 184 of 189 images and useless for triage. Fine-tuning the last layers (stage B) lifts balanced accuracy to 0.562 ± 0.056 and κ to 0.156 ± 0.148; still weak, and with a standard deviation across three seeds that spans zero agreement.

The detector is a different regime. At its deploy threshold it reaches balanced accuracy 0.830 ± 0.115 and $\kappa = 0.677 \pm 0.199$; at OP_{95} it reaches 0.993 ± 0.002 and $\kappa = 0.784 \pm 0.038$. The two-stage arm reaches 0.962 ± 0.059 and $\kappa = 0.818 \pm 0.094$. The high variance of the deploy row is the threshold-transfer problem again: one seed's validation choice produces a conservative test operating point and another's does not.

The MNv3 result is worth stating plainly because it is a negative result about a standard baseline: a classifier that achieves 97.4% accuracy has learned nothing about the task. Accuracy

Table 10: Two-stage ablation on test, mean ± std over detector seeds 0/1/2 (arm thresholds in the second column). Arm A is YOLO at its deploy threshold; B is YOLO at the confidence matching arm A's validation recall; C adds the classifier with protection $P = 0.4$.

arm	conf / τ (s0/s1/s2)	P	R	F_2	FP/img	image R
A	0.02 / 0.01 / 0.21	0.516 ± 0.218	0.976 ± 0.004	0.808 ± 0.102	1.510 ± 1.009	0.993 ± 0.008
B	0.125 / 0.06 / 0.21	0.734 ± 0.050	0.974 ± 0.002	0.913 ± 0.014	0.478 ± 0.127	0.989 ± 0.005
C	τ 0.27 / 0.23 / 0.22	0.814 ± 0.086	0.968 ± 0.004	0.932 ± 0.020	0.309 ± 0.174	0.991 ± 0.003

Table 11: Paired bootstrap for arm C minus arm B, computed separately per detector seed (5,000 resamples over the 189 test images). An interval spanning zero means the arms are indistinguishable at that seed.

detector seed	ΔP [95% CI]	ΔFP/img [95% CI]	ΔR [95% CI]
0	$+0.128$ $[0.065, 0.186]$	-0.2529 $[-0.4603, -0.0952]$	-0.0079 $[-0.0195, 0.0000]$
1	$+0.144$ $[0.096, 0.181]$	-0.3466 $[-0.5979, -0.1587]$	-0.0079 $[-0.0195, 0.0000]$
2	-0.037 $[-0.077, 0.013]$	$+0.0927$ $[-0.0212, 0.2275]$	$+0.0001$ $[-0.0199, 0.0166]$

on a 97.4%-positive test set is not evidence. We keep the row to make the point, and we use the classifier only as a *second stage* over detector crops, where the proposal mechanism supplies the localisation the image-level baseline lacks.

7.4 Two-stage: per-seed, not pooled

Table 10 is the result we most want to be read carefully. Pooled over seeds, arm C looks like a clean win: precision 0.814 ± 0.086 versus 0.734 ± 0.050 for the recall-matched baseline, FP/image 0.31 versus 0.48. The per-seed structure is not a clean win.

Paired bootstrap, one row per detector seed (Table 11).

Seeds 0 and 1 gain precision and lose a small, bounded amount of recall; the precision and false-positive intervals exclude zero. Seed 2 moves the other way: the precision point estimate is negative and every interval spans zero, so the two-stage pipeline does not help that detector.

The mechanism is visible in the arm-A confidences. At seed 2 the validation procedure selects 0.21 for both the deploy and the matched-recall arms, so arms A and B coincide: the baseline is already operating at a well-transferred threshold, and the classifier has no error mass to remove. At seeds 0 and 1 the validation threshold is 0.02 and 0.01, both of which transfer badly to test and produce 1.75 and 2.38 false positives per image. The classifier repairs that specific failure. It is not improving localisation; it is compensating for a threshold that did not generalise.

Two consequences follow, and both cut against the headline. First, the two-stage effect is not a stable property of the architecture; it is conditional on the baseline being badly thresholded. Any paper reporting the three-seed mean without this decomposition is reporting a protocol artifact. Second, if one is allowed to tune the single-stage threshold on validation with the same freedom, the classifier loses under a max-F_2 objective. For seed 0, a YOLO confidence of 0.52 on validation gives test $P = 0.976$, $R = 0.949$, FP/img $= 0.032$, against $P = 0.945$, $R = 0.957$, FP/img $= 0.074$ for the best max-F_2 two-stage configuration. The single stage is 3.1 points better in precision and emits less than half the false alarms, at 0.8 points less recall. The defensible role for the classifier is therefore narrower than the aggregate table suggests: the strict high-recall regime ($R \geq 0.96$) and terrain-nuisance rejection. It is not a general-purpose improvement, and we say so.

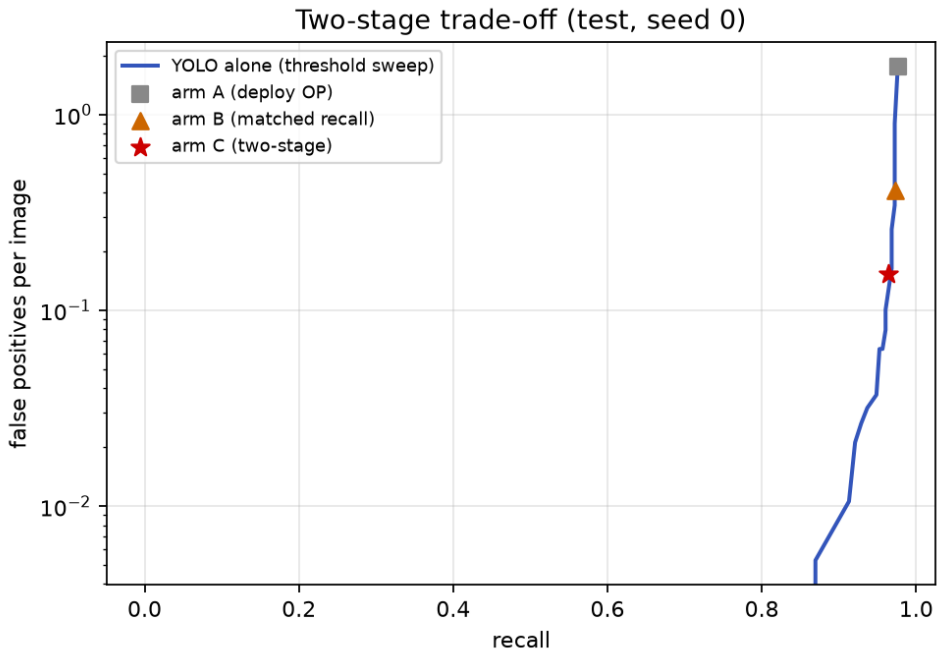


Figure 1: False positives per image against recall for the single-stage and two-stage arms (seed 0). The classifier's advantage is concentrated at high recall; at moderate recall a tuned single-stage threshold is better. Curves are drawn from the same frozen test records used in Table 10.

7.5 Second-stage ablations

Five variants isolate the recipe (Table 12). Removing hard negatives is the largest single degradation ($\Delta P = -0.056$ $[-0.084, -0.028]$), which is unsurprising given that the negatives are detector false positives rather than random background. Removing context padding (0.0) costs 0.027 precision, and doubling it (0.4) costs 0.025: the 20% margin is a genuine optimum rather than a monotone knob. Reducing crop resolution to 160 *improves* precision by 0.017 with an interval that touches zero $[0.000, 0.036]$, a difference we do not claim. Doubling to 320 costs 0.025. All intervals are paired image bootstraps against the baseline on the same test images.

7.6 Grad-CAM: evidence, not correctness

We use attribution as a falsification tool and report what it shows, including where it is uninformative. On 244 true-positive detections, the median share of CAM energy inside the ground-truth box is 0.939 (mean 0.690, first quartile 0.0002, third quartile 0.996). Median enrichment over the box's area share is 58.2×; the peak falls inside the box for 73.0% of detections. The bimodality is important: the distribution is not centred on a high value, it is split between detections whose evidence is almost entirely on the object and detections whose evidence is almost entirely elsewhere. The second group is where the detector is right for the wrong reason.

Deletion tests confirm the direction. Masking the ground-truth box lowers the detection score by 0.739 on average ($n = 182$); masking a matched background region lowers it by 0.0015. The detector's score depends on the object region, as it should.

The head-to-head comparison is the cautionary result. On images where both the detector and the image-level classifier are correct, detector enrichment has median 86.0× and the MobileNetV3

Table 12: Stage-2 ablations at the validation-chosen recall-safe rule, detector seed 0. ΔP is a paired image bootstrap against the baseline; an interval spanning zero means indistinguishable.

variant	pad	crop	hard neg.	τ	P	FP/img	ΔP [95% CI]
baseline	0.2	224	yes	0.27	0.894	0.153	reference
crop 160	0.2	160	yes	0.43	0.911	0.127	$+0.017$ $[0.000, 0.036]$
crop 320	0.2	320	yes	0.08	0.868	0.196	-0.025 $[-0.049, -0.006]$
no hard neg.	0.2	224	no	0.35	0.837	0.254	-0.056 $[-0.084, -0.028]$
pad 0.0	0.0	224	yes	0.16	0.866	0.201	-0.027 $[-0.043, -0.011]$
pad 0.4	0.4	224	yes	0.25	0.868	0.196	-0.025 $[-0.042, -0.011]$

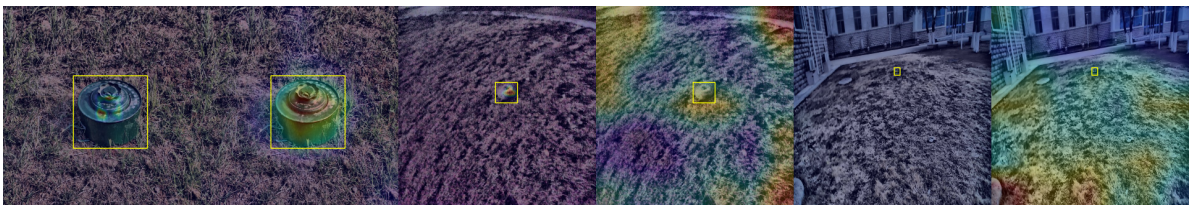


Figure 2: Grad-CAM on detector crops (top) and image-level classifier (bottom). The detector's evidence is object-localised; the classifier's is diffuse. This is a statement about where each model looks, not about whether either prediction is correct.

baseline has median 1.82×; the paired difference is $+147.0$ $[125.4, 170.0]$ in favour of the detector. On detector crops, the stage-2 classifier reaches only 2.13×. The classifier that achieves 97.4% image-level accuracy is not looking at the object in any localised sense; it is exploiting context. That does not make its decisions wrong on this test set, but it explains why fine-tuning it adds so little precision and why its gains vanish when the baseline threshold is already good.

A paste-and-remove test probes the same question behaviourally. Pasting a mine crop into a background yields detection at 0.650 under the deploy confidence and 0.630 at the matched-recall confidence, against 0.976 recall on original images; inpainting a mine out of an image makes it disappear in 100 of 100 composites. The asymmetry is expected: synthetic pastes carry feathered edges and scale mismatch, so the detector is being asked to fire on an object that looks slightly wrong. We report the numbers as an upper bound on the paste setup's realism, not as an estimate of the detector's sensitivity to context. The zero-energy cases are more useful: exactly two test images had zero CAM energy in the box, and both were the objectively wrong labels found in manual review. That independent confirmation is the strongest evidence in this section that the attribution pipeline is measuring something real.

Finally, the failure catalogue (Table 13) is the qualitative counterpart. Of 29 reviewed false positives, 22 are terrain texture; of 9 false negatives, 5 are camouflaged or low-contrast objects and 2 are small. The failure mass is where the deployment risk is, and it is not visible in any aggregate score.

8 Discussion

Report the operating point, then the curve. The size sweep is the cleanest demonstration we have that mAP and deployment behaviour are different measurements. `yolov8n` and `yolov8s` are indistinguishable on test mAP_{50} and differ by 4.3× in false alarms at the recall-first point. The mechanism is not mysterious: mAP integrates over confidence, and a model can rank detections well

Table 13: Hand-reviewed failure categories at the arm-C deployment threshold (seed 0). False positives are capped at 50; 29 were present.

category	FN	FP
camouflaged or low contrast	5	0
small object	2	0
label error	2	0
terrain texture	0	22
low-light ground	0	2
fence / structure edge	0	2
structure wall	0	1
sensor noise	0	1
tree trunk	0	1
total	9	29

while being badly calibrated near the threshold a user will pick. The recommendation is concrete and cheap. Report, at minimum: the validation procedure used to choose a threshold, the threshold itself for every seed, and precision, recall, and false-positives-per-image at that threshold on test. Report mAP as well, for comparability, but do not let it be the only row.

Threshold transfer is the hidden variable. Our three seeds select thresholds of 0.02, 0.01, and 0.21 for the same architecture, same data, same schedule. That is not a bug in the search; it is what happens when a validation set has 181 images and 7 negatives. The consequence is that "model precision" pooled over seeds is mostly a statement about which threshold each seed happened to pick. This also explains the two-stage result: the classifier looks valuable exactly where the threshold transferred badly. Any ablation that changes a component downstream of a fragile threshold will inherit that fragility, and a single pooled mean will hide it.

When is a second stage worth it? On this evidence, narrowly. The classifier is defensible if the requirement is recall ≥ 0.96 with minimum false alarms, and if the detector's threshold is known to be unreliable. It is not defensible as a general precision improvement, because a tuned single-stage threshold beats it at max-F_2, and its benefit disappears at the seed whose threshold transferred well. The cost is real: 16.0 ms per image end to end and 3.06 crops per image on average (95th percentile 3.6). For a pipeline that processes every frame of a flight, that is a meaningful multiplier for a gain that is conditional.

Leakage is worth measuring, not just fixing. The contamination-free comparison is a negative result, and we think negative results of this kind should be routine. Fixing a split is good practice; quantifying what the fix changed is what tells the reader whether the original numbers were wrong by 0.002 or by 0.2. Here it is 0.0003 on the comparable subset. We would not have known that from the audit alone, and we would have been tempted to describe the leak as catastrophic.

What the saliency audit bought. Two things. First, it independently confirmed the two wrong labels, which is a useful cross-check on a manual review. Second, it showed that a 97.4%-accurate image classifier has almost no object-localised evidence. Without that measurement we might have described the MNv3 baseline as "weak but reasonable"; with it, we can say it is exploiting context,

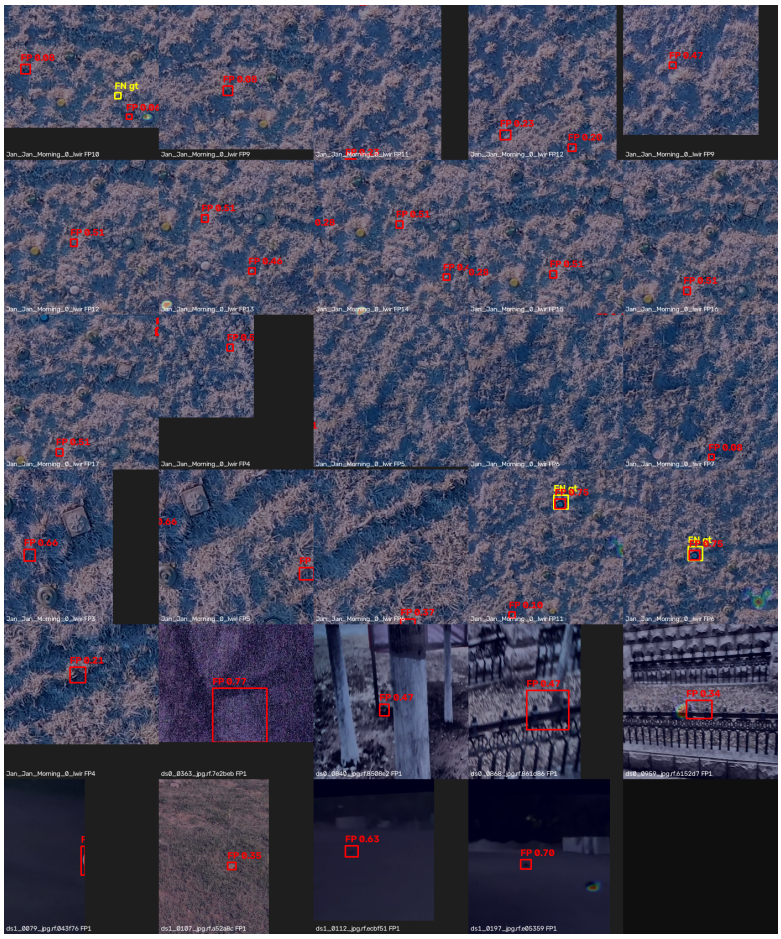


Figure 3: Representative false positives. Terrain texture dominates; the distinction between a mine-like disc and a rock or a patch of ground is the failure mode that matters operationally.

which is a different and more useful statement. Attribution did not prove any prediction correct, and we have tried to keep every sentence on the correct side of that line.

9 Limitations

1. **Compute.** All results come from one 8 GB laptop GPU and roughly 30 GPU-hours. This single constraint explains most of the gaps below. We state it once rather than repeating it as a caveat on every claim.
2. **One detector family, two depths on the clean split.** YOLOv8s (three seeds) and YOLOv8n (one seed). YOLOv8m was trained only on the old leaky split and is therefore not comparable; we exclude it from the size sweep rather than presenting a contaminated point.
3. **Single seed for the small model.** The `yolov8n` row has no interval. Its mAP advantage over `yolov8s` is within the latter's seed spread, and we do not claim it is a better detector.
4. **Five negative test images.** Precision and FP/image are high-variance; the source-cluster intervals span an order of magnitude. Point estimates in tables are accompanied by intervals wherever the interval is meaningful.

5. **Label correction without retraining.** Two test boxes were removed from ground truth. We report both versions, but the model was not retrained on corrected labels because no training-label errors were found.
6. **No grouped cross-validation.** A single frozen split was used, as required by the protocol. Group-aware k-fold would give a better estimate of split variance and was not affordable.
7. **No hard-negative re-mining on unseen data.** The stage-2 negatives come from detector false positives on the training split, which is the correct procedure but limits how novel the negatives are.
8. **Synthetic composites.** The paste-and-remove experiment uses feathered, scale-mismatched composites. It is a probe, not a counterfactual.
9. **Not edge-validated.** All latency numbers are PC GPU FP32. No Jetson, TensorRT, or flight hardware was used; the edge phases were dropped from scope.
10. **Source confound.** `Jan` and `May` are colour photographs with marker pins, despite `lwir` in the filename. Marker shortcuts cannot be fully excluded, and the source dominates aggregate false alarms.
11. **No field validation and no buried-object claim.** This is a benchmark on visible surface indicators. It says nothing about burial depth, soil type, or detection probability in real clearance conditions.

10 Ethics and Dual Use

What this system can and cannot do. The detector recognises visible mine-like objects and surface indicators in downward-looking colour imagery. It cannot detect a buried object, because the sensor cannot see one. It has not been flown, field-tested, or validated against ground truth collected in a real clearance operation. Nothing in this paper supports a statement about detection probability in the field, and we do not make one. The correct description is a computer-vision benchmark on a public, publicly derived image corpus.

Dual use. The same capability that helps a clearance team prioritise ground could help someone avoid marked hazards. We publish standard detection metrics, an audit methodology, and trained weights; we publish no targeting guidance, no geographic coordinates, and no operational deployment procedure. The dual-use risk is real but generic to visible-object detection, and we judge that the methodological contribution—how to audit a leaky split and why to report operating points—outweighs it. We note that the weights are released under AGPL-3.0, which requires derivative services to publish their source; this is a licence choice, not an ethical safeguard.

The cost of the miss rate. At the primary operating point the two-stage arm recalls 0.964 of annotated instances, which on this test set means nine missed objects out of 253. Those nine are analysed in Table 13: five are camouflaged or low-contrast, two are small objects, and two are the objectively wrong labels removed under D023. A recall of 0.964 reads as a good number in a table. In a clearance context it means that roughly one in twenty-eight annotated objects is not flagged, and the misses are concentrated in exactly the low-contrast cases where a human reviewer is most likely to defer to the model. No metric in this paper should be read as saying the system is safe to rely on.

Data ethics. The corpus is derived from public dataset exports and contains no human subjects, no personal data, and no imagery of identified individuals. We did not collect new data. The original licences were not recorded at download time; we therefore redistribute no images (Section 3). Where figures include source imagery, they are reproduced for scholarly comment at low resolution and are not a dataset release.

Reporting. We report the results that weaken our own claims—the seed-2 failure of the two-stage pipeline, the negative leakage result, the degenerate image-level baseline, the tuned single-stage threshold that beats the pipeline—in the main text rather than in an appendix. The reason is practical: a reader who is deciding whether to trust the positive results needs to know how the negative ones were handled.

11 Availability and Licensing

Code and artifacts. Training, evaluation, auditing, Grad-CAM and table-generation code are released with the paper, together with all evaluation JSONs, the generated tables, and the figures. The frozen split manifest is released as a CSV with the following SHA-256 digest:

`748322d62ace769445d0bd2363699097`
`0c7a5115a99c9344e139099c608c80de`

No image, label file, or crop is redistributed; see `DATA_PROVENANCE.md` for the reason and for the exact source inventory.

Weights. Three detector seeds are released, each with `best.pt`, `last.pt` and an ONNX export, plus three stage-2 checkpoints and the original v1 run for reference, all with a SHA-256 manifest. The detector checkpoints are derivative works of Ultralytics YOLOv8 and are distributed under **AGPL-3.0**; the stage-2 checkpoints are fine-tuned torchvision MobileNetV3-Large and are distributed under the same repository licence for consistency. Third-party terms are stated in `models/NOTICE.md`. The repository code and the audit artifacts are AGPL-3.0.

Reproduction. Regenerating every table requires no GPU and no dataset: the committed JSONs are sufficient. The table script exits with an error if a required field is absent rather than substituting a value, so a clean clone either reproduces the tables in this paper or fails visibly.

Persistent identifier. [TODO: assign DOI via Zenodo on release; insert the DOI and the exact release tag here. The manifest digest above is the content-level identifier that will not change.]

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A Decision log (abridged)

The project kept a numbered decision register. The entries that bear on claims in this paper are reproduced here; the full register, with dates and rationale for all 23 entries, is in `DECISIONS.md`.

Table 14: Abridged decision register. Entries omitted here concern file layout and tooling rather than claims.

ID	Decision and reason
D010	Reject the original split. 76 cross-split base-name groups; 63.8% of evaluation images with a pHash-≤ 2 training neighbour; augmentation before splitting confirmed in the upstream scripts.
D011	Image-level and FP/image metrics blocked until the test split contains negatives. It contained zero.
D012	`May`/`Jan` files are colour RGB with marker pins, not thermal, despite the `lwir` filename. Keep and disclose (D016).
D014	Operational leak definition: exact MD5, or base key, or same-source pHash≤ 2 *and* 32×32 RGB MSE ≤ 900. Manual review of all 146 candidates sets the MSE cut (same-scene ≤ 817, texture ≥ 958.6).
D015	Freeze `landmine_combined_v2` at 3,907/181/189 with 30 eval leaks removed; manifest digest `748322d6...c80de`; true leak rate 0/370.
D017	Disclosed protocol deviation: the upstream training script auto-evaluates test at confidence 0.25. No selection uses those numbers; the frozen-protocol evaluation is a separate single pass.
D018	Accept early stopping at epoch 67/75 (best 47) for the pilot run; retrain at batch 8 for comparability (D019). The pilot is not a paper result.
D020	Frozen-feature MNv3 is degenerate on negatives at every resolution and batch size ($\kappa \approx 0$); only fine-tuning the last layers separates them. Stage A rows retained as the mandated ablation.
D021	Threshold rule: sweep protection P, choose the largest τ within 0.5 points of stage-1 validation recall, then minimise validation FP/image. All seeds select $P = 0.4$.
D022	Grad-CAM flags images by energy in the union of all ground-truth boxes, not per detection; per-detection rows were an instance-count artefact on multi-mine frames.
D023	Remove two objectively wrong test boxes (tree trunk, fence); report before/after; do not retrain (no training-label errors found).

B Test-touch accounting

Eighteen numbered test touches were logged. Grouped by purpose:

Table 15: Test-touch accounting, grouped by purpose.

touches	purpose	sanctioning rule
#1–#2	audit read-only passes	leakage audit reads labels only
#3	MNv3 baseline, seed 0	Phase 4 protocol eval, val-chosen thresholds
#4	YOLO seed-0 frozen-protocol eval	the single protocol-governing eval
#5	MNv3 seeds 1–2	Phase 4 protocol eval
#6–#8	two-stage seeds 0–2	Phase 5 protocol eval, val-frozen
#9–#12	Grad-CAM E1–E6	Phase 6, read-only attribution
#13–#14	YOLO seeds 1–2	Phase 3 per-seed protocol eval
#15–#16	leaky baseline, v1 split and clean subset	extended analysis
#17	YOLOv8n size point	size ablation, val-chosen threshold
#18	corrected detector-aligned two-stage	re-score, no selection

The only touch that could contaminate a claim is #4, and it is the evaluation the paper's detector numbers come from. Touch #18 is a correction: the original seed-1/2 two-stage runs had reused

the seed-0 detector, which we detected from byte-identical arm-A/arm-B records and corrected by re-running with each seed's own checkpoint. No threshold was re-chosen.

C Failure gallery and reproduction commands

The false-positive contact sheet is reproduced in Figure 3. The full false-negative gallery, the audit montages, the per-image predictions, and the Grad-CAM heatmaps are in the repository under `paper_results/`.

```
# tables T1-T6 and figures (no GPU, no dataset required)
python src/reporting/make_tables.py

# frozen-protocol evaluation of a checkpoint
python src/evaluation/phase3_eval.py --run <run_dir> --out <out_dir>

# two-stage pipeline with each detector seed
python src/pipeline/phase5_pipeline.py --yolo <seed_best.pt> \
    --cls <stage2_best.pt> --arm-a-conf <val_conf> --out <out_dir>

# statistical analysis, bootstrap and policy rules
python src/analysis/analysis_stats.py
```

Environment: Python 3.11.9, Ultralytics 8.4.117, PyTorch 2.6.0+cu124, torchvision 0.21.0+cu124, NVIDIA RTX 4070 Laptop (8 GB). Evaluation at 640×640, NMS IoU 0.7, `max_det` 300, FP32, no TTA.